

ASSIGNMENT 14.1

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Batch-27

QUESTION-1

You are building a personal portfolio website to showcase your work.

Requirements:

- Create sections for About Me, Projects, and Contact.
- Use AI to:
 - o Suggest color palettes and typography.
 - o Create a responsive layout with Grid/Flexbox.
 - o Add smooth scrolling navigation

CODE:

```

index.html > html > head > style > html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Personal Portfolio</title>
7 <!-- AI-suggested color palette and typography -->
8 <style>
9 :root {
10   --primary: #2D3142;
11   --accent: #EF8354;
12   --background: #F5F5F5;
13   --text: #4F5D75;
14   --heading: #1A1A2E;
15   --font-main: 'Montserrat', Arial, sans-serif;
16   --font-heading: 'Poppins', Arial, sans-serif;
17 }
18 html {
19   scroll-behavior: smooth;
20   background: var(--background);
21   color: var(--text);
22   font-family: var(--font-main);
23 }
24 body {
25   margin: 0;
26   padding: 0;
27 }
28 header {
29   background: var(--primary);
30   color: #fff;
31   padding: 1rem 0;
32   position: sticky;
33   top: 0;
34   z-index: 100;
35 }
36 nav {
37   display: flex;

```

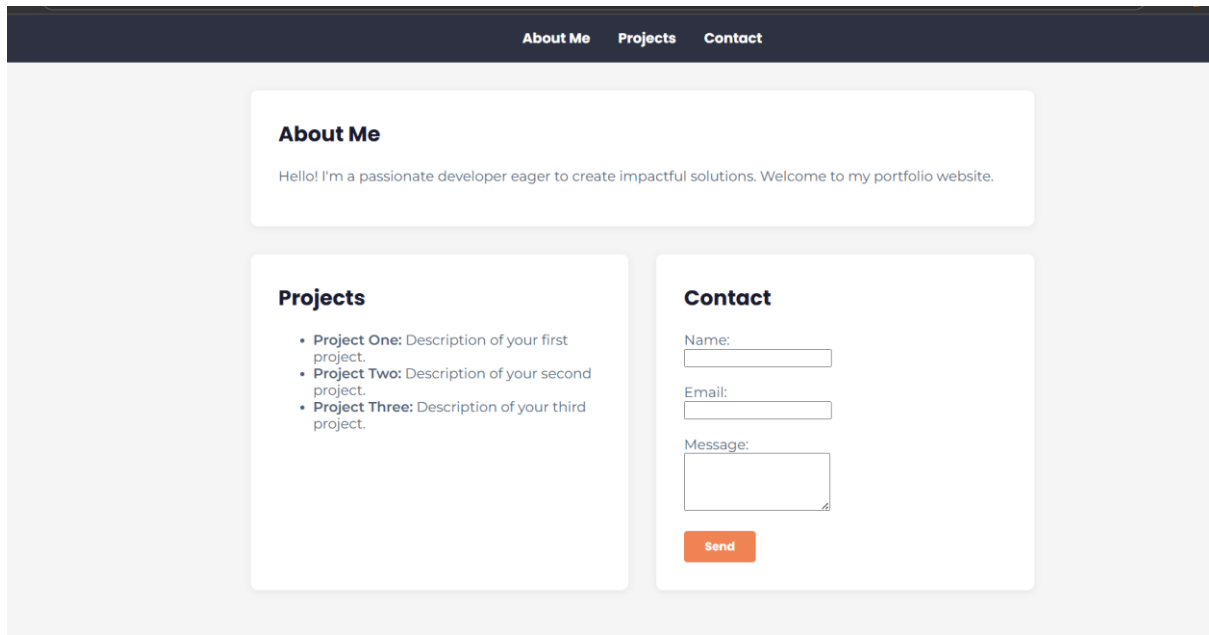
```

index.html x
index.html > html > head > style > html
2 <html lang="en">
3 <head>
8 <style>
36   nav {
38     justify-content: center;
39     gap: 2rem;
40   }
41   nav a {
42     color: #fff;
43     text-decoration: none;
44     font-weight: 600;
45     font-family: var(--font-heading);
46     transition: color 0.3s;
47   }
48   nav a:hover {
49     color: var(--accent);
50   }
51   main {
52     display: grid;
53     grid-template-columns: 1fr;
54     gap: 2rem;
55     max-width: 900px;
56     margin: 2rem auto;
57     padding: 0 1rem;
58   }
59   section {
60     background: #fff;
61     border-radius: 8px;
62     box-shadow: 0 2px 8px rgba(45,49,66,0.08);
63     padding: 2rem;
64   }
65   h2 {
66     font-family: var(--font-heading);
67     color: var(--heading);
68     margin-top: 0;
69   }
70   /* Responsive layout */

```

```
index.html X
index.html > html > head > style > html
2  <html lang="en">
3  <head>
8  <style>
70  /* Responsive layout */
71  @media (min-width: 700px) {
72    main {
73      grid-template-columns: repeat(2, 1fr);
74    }
75    #about {
76      grid-column: 1 / 3;
77    }
78  }
79  @media (max-width: 699px) {
80    main {
81      display: flex;
82      flex-direction: column;
83    }
84  }
85 </style>
86 <!-- Google Fonts -->
87 <link href="https://fonts.googleapis.com/css?family=Montserrat:400,600&display=swap" rel="stylesheet">
88 <link href="https://fonts.googleapis.com/css?family=Poppins:700&display=swap" rel="stylesheet">
89 </head>
90 <body>
91   <header>
92     <nav>
93       <a href="#about">About Me</a>
94       <a href="#projects">Projects</a>
95       <a href="#contact">Contact</a>
96     </nav>
97   </header>
98   <main>
99     <section id="about">
100       <h2>About Me</h2>
101       <p>Hello! I'm a passionate developer eager to create impactful solutions. Welcome to my portfolio website.</p>
102     </section>
103     <section id="projects">
104       <h2>Projects</h2>
105       <ul>
106         <li><strong>Project One:</strong> Description of your first project.</li>
107         <li><strong>Project Two:</strong> Description of your second project.</li>
108         <li><strong>Project Three:</strong> Description of your third project.</li>
109       </ul>
110     </section>
111     <section id="contact">
112       <h2>Contact</h2>
113       <form>
114         <label for="name">Name:</label><br>
115         <input type="text" id="name" name="name" required><br><br>
116         <label for="email">Email:</label><br>
117         <input type="email" id="email" name="email" required><br><br>
118         <label for="message">Message:</label><br>
119         <textarea id="message" name="message" rows="4" required></textarea><br><br>
120         <button type="submit" style="background: var(--accent); color: #fff; border: none; padding: 0.5rem 1.5rem; border-radius: 0.5rem;>Submit</button>
121       </form>
122     </section>
123   </main>
124 </body>
125 </html>
126
```

OUTPUT:



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QUESTION 2

Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

- Display product image, title, price, and "Add to Cart" button.
- Use AI to:
 - o Style with BEM methodology.
 - o Make layout responsive.
 - o Add hover effects and "Add to Cart" alert

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CODE

```
index2.html > ...
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Online Store Product Page</title>
7   <style>
8     /* BEM methodology */
9     .product-page {
10       display: flex;
11       justify-content: center;
12       align-items: center;
13       min-height: 100vh;
14       background: #f5f5f5;
15     }
16     .product-page__container {
17       background: #fff;
18       border-radius: 12px;
19       box-shadow: 0 4px 16px rgba(0,0,0,0.08);
20       padding: 32px;
21       max-width: 400px;
22       width: 100%;
23       display: flex;
24       flex-direction: column;
25       align-items: center;
26     }
27     .product-page__image {
28       width: 100%;
29       max-width: 260px;
30       border-radius: 8px;
31       margin-bottom: 24px;
32       transition: box-shadow 0.3s;
33     }
34     .product-page__image:hover {
35       box-shadow: 0 8px 24px rgba(0,0,0,0.15);
36     }
```

```

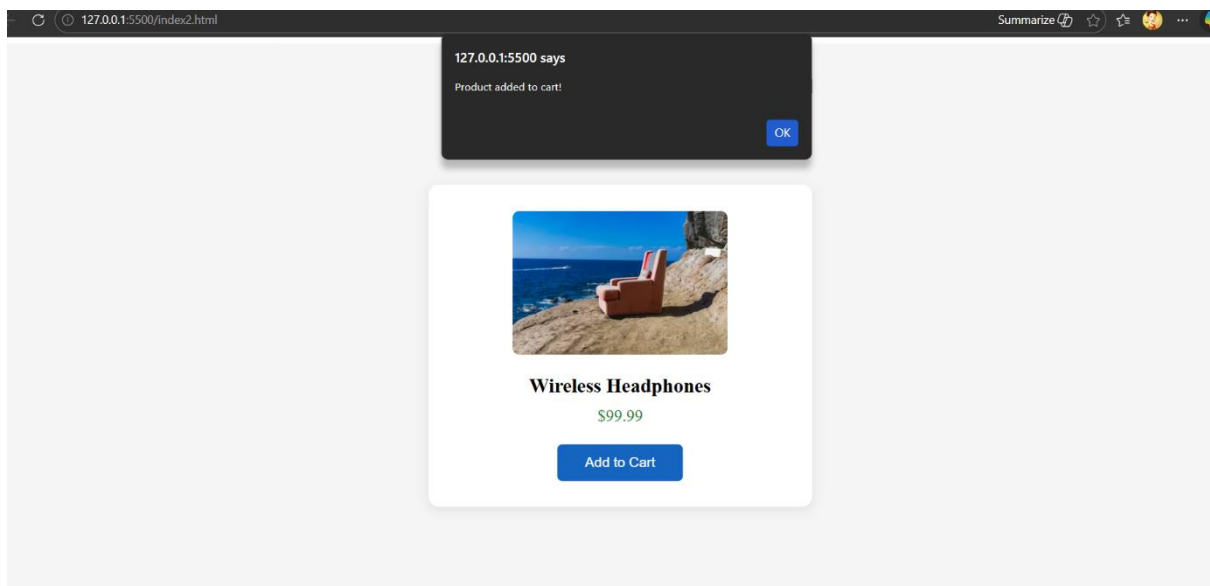
37     .product-page__title {
38       font-size: 1.5rem;
39       font-weight: 600;
40       margin-bottom: 12px;
41       text-align: center;
42     }
43     .product-page__price {
44       font-size: 1.25rem;
45       color: #2e7d32;
46       margin-bottom: 24px;
47     }
48     .product-page__add-to-cart {
49       background: #1976d2;
50       color: #fff;
51       border: none;
52       border-radius: 6px;
53       padding: 12px 32px;
54       font-size: 1rem;
55       font-weight: 500;
56       cursor: pointer;
57       transition: background 0.3s, transform 0.2s;
58     }
59     .product-page__add-to-cart:hover {
60       background: #1565c0;
61       transform: scale(1.05);
62     }
63     @media (max-width: 600px) {
64       .product-page__container {
65         padding: 16px;
66         max-width: 100%;
67       }
68       .product-page__image {
69         max-width: 100%;
70       }
71     }
```

```

<head>
  <style>
    @media (max-width: 600px) {
      .product-page__container {
        max-width: 100%;
      }
      .product-page__image {
        max-width: 100%;
      }
    }
  </style>
</head>
<body>
  <div class="product-page">
    <div class="product-page__container">
      
      <div class="product-page__title">Wireless Headphones</div>
      <div class="product-page__price">$99.99</div>
      <button class="product-page__add-to-cart" onclick="addToCartAlert()">Add to Cart</button>
    </div>
  </div>
  <script>
    function addToCartAlert() {
      alert('Product added to cart!');
    }
  </script>
</body>
</html>

```

OUTPUT:



QUESTION 3

Code:

```
name.addEventListener('input', validateName);
email.addEventListener('input', validateEmail);
phone.addEventListener('input', validatePhone);
session.addEventListener('change', validateSession);

form.addEventListener('submit', function(e) {
    let valid = true;
    if (!validateName()) valid = false;
    if (!validateEmail()) valid = false;
    if (!validatePhone()) valid = false;
    if (!validateSession()) valid = false;
    if (!valid) {
        e.preventDefault();
    }
});
</script>
</body>
</html>
```

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Conference Event Registration</title>
7   <style>
8     body {
9       font-family: Arial, sans-serif;
10      background: #f4f6f8;
11      margin: 0;
12      padding: 0;
13    }
14    .container {
15      max-width: 420px;
16      margin: 40px auto;
17      background: #fff;
18      border-radius: 8px;
19      box-shadow: 0 2px 8px rgba(0,0,0,0.08);
20      padding: 32px 24px;
21    }
22    h2 {
23      text-align: center;
24      margin-bottom: 24px;
25      color: #2c3e50;
26    }
27    label {
28      display: block;
29      margin-bottom: 6px;
30      font-weight: 500;
31    }
32    input, select {
33      width: 100%;
34      padding: 10px;
35      margin-bottom: 18px;
36      border: 1px solid #ccd0d5;
37      border-radius: 4px;
```

```

    input, select {
      font-size: 16px;
    }
    input:focus, select:focus {
      border-color: #2980b9;
      outline: none;
    }
    .error {
      color: #e74c3c;
      font-size: 14px;
      margin-top: -14px;
      margin-bottom: 10px;
    }
    button {
      width: 100%;
      padding: 12px;
      background: #2980b9;
      color: #fff;
      border: none;
      border-radius: 4px;
      font-size: 18px;
      font-weight: 600;
      cursor: pointer;
      transition: background 0.2s;
    }
    button:hover {
      background: #2471a3;
    }
  }
</style>
</head>
<body>
  <div class="container">
    <h2>Conference Registration</h2>
    <form id="registrationForm" aria-labelledby="formTitle" novalidate>

```

```

    <label for="name" aria-label="Full Name">Full Name</label>
    <input type="text" id="name" name="name" required aria-required="true" aria-describedby="nameError">
    <div id="nameError" class="error" aria-live="polite"></div>

    <label for="email" aria-label="Email Address">Email Address</label>
    <input type="email" id="email" name="email" required aria-required="true" aria-describedby="emailError">
    <div id="emailError" class="error" aria-live="polite"></div>

    <label for="phone" aria-label="Phone Number">Phone Number</label>
    <input type="tel" id="phone" name="phone" required aria-required="true" aria-describedby="phoneError" pattern="[0-9]{10}">
    <div id="phoneError" class="error" aria-live="polite"></div>

    <label for="session" aria-label="Session Selection">Session Selection</label>
    <select id="session" name="session" required aria-required="true" aria-describedby="sessionError">
      <option value="">-- Select a session --</option>
      <option value="morning">Morning Session</option>
      <option value="afternoon">Afternoon Session</option>
      <option value="evening">Evening Session</option>
    </select>
    <div id="sessionError" class="error" aria-live="polite"></div>

    <button type="submit" aria-label="Submit Registration">Register</button>
  </form>
</div>
<script>
  const form = document.getElementById('registrationForm');
  const name = document.getElementById('name');
  const email = document.getElementById('email');
  const phone = document.getElementById('phone');
  const session = document.getElementById('session');
  const nameError = document.getElementById('nameError');
  const emailError = document.getElementById('emailError');
  const phoneError = document.getElementById('phoneError');

```



```

</div>
<script>
const form = document.getElementById('registrationForm');
const name = document.getElementById('name');
const email = document.getElementById('email');
const phone = document.getElementById('phone');
const session = document.getElementById('session');
const nameError = document.getElementById('nameError');
const emailError = document.getElementById('emailError');
const phoneError = document.getElementById('phoneError');
const sessionError = document.getElementById('sessionError');

function validateName() {
  if (!name.value.trim()) {
    nameError.textContent = 'Name is required.';
    return false;
  }
  nameError.textContent = '';
  return true;
}

function validateEmail() {
  const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
  if (!email.value.trim()) {
    emailError.textContent = 'Email is required.';
    return false;
  } else if (!emailPattern.test(email.value)) {
    emailError.textContent = 'Enter a valid email address.';
    return false;
  }
  emailError.textContent = '';
  return true;
}

function validatePhone() {
  const phonePattern = /^[0-9\-\+\s]{7,15}$/;
  if (!phone.value.trim()) {
    phoneError.textContent = 'Phone number is required.';
    return false;
  } else if (!phonePattern.test(phone.value)) {
    phoneError.textContent = 'Enter a valid phone number.';
    return false;
  }
  phoneError.textContent = '';
  return true;
}

function validateSession() {
  if (!session.value) {
    sessionError.textContent = 'Please select a session.';
    return false;
  }
  sessionError.textContent = '';
  return true;
}

name.addEventListener('input', validateName);
email.addEventListener('input', validateEmail);
phone.addEventListener('input', validatePhone);
session.addEventListener('change', validateSession);

form.addEventListener('submit', function(e) {
  let valid = true;
  if (!validateName()) valid = false;
  if (!validateEmail()) valid = false;
  if (!validatePhone()) valid = false;
  if (!validateSession()) valid = false;
  if (!valid) {

```

Output:

Conference Registration

Full Name

Email Address

Phone Number

Session Selection

-- Select a session --

Register

QUESTION 4)

CODE:

```
> index.html > -
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Resume Website</title>
7    <style>
8      /* Professional Typography */
9      body {
10        font-family: 'Segoe UI', 'Roboto', Arial, sans-serif;
11        font-size: 18px;
12        color: #222;
13        background: #f7f7f7;
14        margin: 0;
15        line-height: 1.6;
16      }
17      h1, h2 {
18        font-family: 'Montserrat', 'Segoe UI', Arial, sans-serif;
19        font-weight: 700;
20        letter-spacing: 0.05em;
21        margin-bottom: 0.5em;
22      }
23      h1 {
24        font-size: 2.5rem;
25        color: #2a2a2a;
26      }
27      h2 {
28        font-size: 1.5rem;
29        color: #3a3a3a;
30      }
31      p, li {
32        font-size: 1.1rem;
33      }
34      /* Smooth Scroll */
35      html {
36        scroll-behavior: smooth;
37      }
38
39      /* Navigation */
40      nav {
41        background: #fff;
42        box-shadow: 0 2px 8px rgba(0,0,0,0.07);
43        padding: 1em 2em;
44        position: sticky;
45        top: 0;
46        z-index: 10;
47        display: flex;
48        justify-content: center;
49      }
50      nav a {
51        color: #2a2a2a;
52        text-decoration: none;
53        margin: 0 1.5em;
54        font-weight: 600;
55        transition: color 0.2s;
56      }
57      nav a:hover {
58        color: #0077b6;
59      }
60
61      /* CSS Grid Layout */
62      .container {
63        display: grid;
64        grid-template-columns: 1fr;
65        gap: 2.5em;
66        max-width: 900px;
67        margin: 3em auto;
68        padding: 2em;
69        background: #fff;
70        border-radius: 12px;
71        box-shadow: 0 4px 24px rgba(0,0,0,0.09);
72      }
```

```

7 <style>
1   section {
2     padding-bottom: 1em;
3     border-bottom: 1px solid #ecec;
4   }
5   section:last-child {
6     border-bottom: none;
7   }
8   ul {
9     padding-left: 1.5em;
10  }
11  @media (max-width: 600px) {
12    .container {
13      padding: 1em;
14    }
15    nav {
16      padding: 0.5em 1em;
17    }
18    h1 {
19      font-size: 1.7rem;
20    }
21    h2 {
22      font-size: 1.1rem;
23    }
24  }
25 </style>
26 </head>
27 <body>
28   <nav>
29     <a href="#profile">Profile Summary</a>
30     <a href="#skills">Skills</a>
31     <a href="#education">Education</a>
32     <a href="#experience">Experience</a>
33   </nav>
34   <div class="container">

```

```

<section id="profile">
  <h1>Profile Summary</h1>
  <p>Dynamic and results-driven professional with a passion for technology and innovation. Experienced in delivering
</section>
<section id="skills">
  <h2>Skills</h2>
  <ul>
    <li>Web Development (HTML, CSS, JavaScript)</li>
    <li>Python & Data Analysis</li>
    <li>Project Management</li>
    <li>UI/UX Design</li>
    <li>Problem Solving</li>
  </ul>
</section>
<section id="education">
  <h2>Education</h2>
  <ul>
    <li>B.Sc. in Computer Science, XYZ University (2018-2022)</li>
    <li>Certified Scrum Master (2023)</li>
  </ul>
</section>
<section id="experience">
  <h2>Experience</h2>
  <ul>
    <li>Software Developer, ABC Tech (2022-Present)</li>
    <li>Intern, DEF Solutions (2021-2022)</li>
  </ul>
</section>
</div>
</body>
</html>

```

OUTPUT:

Profile SummarySkillsEducationExperience

Profile Summary

Dynamic and results-driven professional with a passion for technology and innovation. Experienced in delivering high-quality solutions and collaborating with cross-functional teams to achieve business goals.

Skills

- Web Development (HTML, CSS, JavaScript)
- Python & Data Analysis
- Project Management
- UI/UX Design
- Problem Solving

QUESTION 5)

CODE:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Restaurant Menu</title>
7   <style>
8     /* AI-suggested food-friendly color palette */
9     :root {
10       --primary: #e67e22; /* carrot orange */
11       --secondary: #f6e2b3; /* light beige */
12       --accent: #27ae60; /* fresh green */
13       --card-bg: #fffaf3; /* soft white */
14       --text-main: #2d3436; /* dark gray */
15       --hover-bg: #ffe5b4; /* peach */
16     }
17     body {
18       background: var(--secondary);
19       font-family: 'Segoe UI', Arial, sans-serif;
20       color: var(--text-main);
21       margin: 0;
22       padding: 0;
23     }
24     .menu-container {
25       display: flex;
26       flex-wrap: wrap;
27       gap: 2rem;
28       justify-content: center;
29       padding: 2rem;
30     }
31     .menu-card {
32       background: var(--card-bg);
33       border-radius: 16px;
34       box-shadow: 0 4px 16px #2d3436;
35       width: 320px;
36       max-width: 90vw;
37       padding: 1.5rem;
38     }
39   </style>
40 </html>
```

```
41     .menu-card {
42       transition: transform 0.2s, box-shadow 0.2s, background 0.2s;
43       display: flex;
44       flex-direction: column;
45       align-items: flex-start;
46     }
47     .menu-card:hover {
48       transform: translateY(-8px) scale(1.03);
49       box-shadow: 0 8px 24px #2d3436;
50       background: var(--hover-bg);
51     }
52     .menu-title {
53       font-size: 1.4rem;
54       font-weight: bold;
55       color: var(--primary);
56       margin-bottom: 0.5rem;
57     }
58     .menu-desc {
59       font-size: 1rem;
60       margin-bottom: 1rem;
61       color: var(--text-main);
62     }
63     .menu-price {
64       font-size: 1.1rem;
65       font-weight: 600;
66       color: var(--accent);
67       margin-top: auto;
68     }
69     @media (max-width: 700px) {
70       .menu-container {
71         gap: 1rem;
72         padding: 1rem;
73       }
74       .menu-card {
75         width: 100%;
76         max-width: none;
77       }
78     }
79   </style>
80 </html>
```

```

padding: 1rem;
    }
  }
</style>
</head>
<body>
  <h1 style="text-align:center;color:var(--primary);margin-top:2rem;">Our Menu</h1>
  <div class="menu-container">
    <div class="menu-card">
      <div class="menu-title">Margherita Pizza</div>
      <div class="menu-desc">Classic pizza topped with fresh tomatoes, mozzarella, basil, and olive oil.</div>
      <div class="menu-price">$12.99</div>
    </div>
    <div class="menu-card">
      <div class="menu-title">Grilled Salmon</div>
      <div class="menu-desc">Salmon fillet grilled to perfection, served with lemon butter sauce and seasonal vegetables.</div>
      <div class="menu-price">$18.50</div>
    </div>
    <div class="menu-card">
      <div class="menu-title">Caesar Salad</div>
      <div class="menu-desc">Crisp romaine lettuce, parmesan, croutons, and creamy Caesar dressing.</div>
      <div class="menu-price">$9.00</div>
    </div>
    <div class="menu-card">
      <div class="menu-title">Spaghetti Carbonara</div>
      <div class="menu-desc">Traditional Italian pasta with pancetta, egg, parmesan, and black pepper.</div>
      <div class="menu-price">$13.75</div>
    </div>
    <div class="menu-card">
      <div class="menu-title">Chocolate Lava Cake</div>
      <div class="menu-desc">Warm chocolate cake with a gooey molten center, served with vanilla ice cream.</div>
      <div class="menu-price">$7.50</div>
    </div>
  </div>

```

